

Univariate:

1) Central Decency

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[11]: for colName in quan:
        descriptive[colName]["Mean"] = dataset[colName].mean()
        descriptive[colName]["Median"] = dataset[colName].median()
        descriptive[colName]["Mode"] = dataset[colName].mode()[0]
descriptive
```

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288658.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0

In this image, Means almost close for Sl_no, ssc_p, degress_p, etest_p, mba_p. but salary only having high level difference in mean and median. So, salary having outlier.

2) Percentile

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17]: for colName in quan:
        descriptive[colName]["Mean"] = dataset[colName].mean()
        descriptive[colName]["Median"] = dataset[colName].median()
        descriptive[colName]["Mode"] = dataset[colName].mode()[0]
        descriptive[colName]["Q1:25%"] = dataset.describe()[colName]["25%"]
        descriptive[colName]["Q2:50%"] = dataset.describe()[colName]["50%"]
        descriptive[colName]["Q3:75%"] = dataset.describe()[colName]["75%"]
        descriptive[colName]["99%"] = np.percentile(dataset[colName],99)
        descriptive[colName]["Max"] = dataset.describe()[colName]["max"]
descriptive
```

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17]:
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	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Max	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

```
12]: import numpy as np
```

In this image, Q1, Q2, Q3, 99%, max are different type of location for splitting data percentage.

For Examples, Consider as ssc_p

Q1 = 25% , Q2= 50% = 60.6 -67.0 = overall 7% percentage difference from Q1 to Q2

Q2 = 50% , Q3= 75% = 67.0 -75.7 = overall 8% percentage difference from Q2 to Q3

Q3 = 50% , 99% = 75.7-87.0 = overall 12% percentage difference from Q3 to 99%

99% , Q4= 100% or MAX = 87.0 -89.4 = overall 2% percentage difference from 99% to 100%

Based on above data each percentage wise categorize the data percentage.

3. Find the IQR:

- A) The InterQuartile range. Compare the interquartile ranges
- B) Any outliers in either set.

The five number summary for the day and night class is

a. The interquartile range. Compare the two interquartile ranges.					
b. Any outliers in either set.					
The five number summary for the day and night classes is					
	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

A) $IQR = Q_3 - Q_1$

$$IQR \text{ for day} = 82.5 - 56 = 26.5$$

$$IQR \text{ for night} = 89 - 78 = 11$$

Difference between night and day $IQR = 15.5$

- B) less Outlier = $Q_1 - 1.5 * IQR$
- greater Outlier = $Q_3 + 1.5 * IQR$

Day outlier:

$$\text{Less outlier} = 56 - 1.5 * 26.5 = 16.25$$

$$\text{Greater Outlier} = 89 + 1.5 * 26.5 = 128.75$$

Day – There is not less or greater outlier

Night outlier:

$$\text{Less outlier} = 78 - 1.5 * 11 = 61.5$$

$$\text{Greater Outlier} = 89 + 1.5 * 11 = 105.5$$

night – There is having less outlier because the minimum values is 25.5. But less outlier should be 61.50. it is 25.5 is less than 61.5. The greater outlier is not available in this night dataset.

Find out Outlier:

[8]:	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5Rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Greater	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0

Sl_no

Min = 1, Max = 215

Less Outlier = -106.0, Greater Outlier = 322.0

There is no less or greater outlier.

ssc_p

Min = 40.89, Max = 89.4

Less Outlier = 37.95, Greater Outlier = 98.35

There is no less or greater outlier.

Hsc_p

Min = 37.0, Max = 97.7

Less Outlier = 42.75, Greater Outlier = 91.15

There is having less and greater outlier.

degree_p

Min =50.0, Max = 91.0

Less Outlier =44.5, Greater Outlier = 88.5

There is no less outlier and greater outlier.

Etest_p

Min =50.0, Max = 98.0

Less Outlier =24.75, Greater Outlier = 118.75

There is no less outlier and greater outlier.

Mba_p

Min =51.21, Max = 77.89

Less Outlier =45.48, Greater Outlier = 78.72

There is no less outlier and greater outlier.

Salary:

Min =200000, Max = 940000

Less Outlier =150000, Greater Outlier = 390000

There is no less outlier and having greater outlier.

4. Skewness:

- Skew = 0 → symmetric

Short ← Average → Tall

Most people are around average height, with similar numbers of shorter and taller people.

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- Skew > 0 → right-skewed (tail on the right, e.g., income data)

Many people Few people

earning less earning much more

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- Skew < 0 → left-skewed (tail on the left, e.g., age at retirement)

Most people retire around 60- 65 years, but a small number retire much earlier due to health or personal

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Sl_No = 0 - symmetric

Ssc_p = -0.132649 - left-skewed

Hsc_p = 0.162611 – Right Skewed

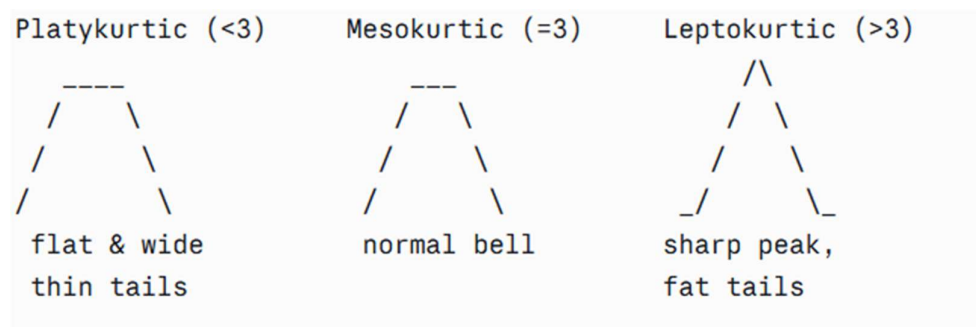
Degree_p = 0.204164 - Right Skewed

Etest_p = 0.282308- Right Skewed

Mba_p = 0.313576 - Right Skewed

Salary = 0.8067 - Right Skewed

5) Kurtosis



1. Mesokurtic (Kurtosis = 3)

Real-Life Example: Adult Heights

Most adults are close to the average height.

- Few people are extremely short.
- Few people are extremely tall.
- Most people are around the middle.

2. Leptokurtic (Kurtosis > 3)

Real-Life Example: Company Salaries

Imagine a company with 1,000 employees:

- Most employees earn between \$55,000 and \$65,000.
- A few executives earn \$500,000+.

3. Platykurtic (Kurtosis < 3)

Real-Life Example: Dice Rolls

Roll a fair dice many times.

Mesokurtic (Kurtosis = 3)

Leptokurtic (Kurtosis > 3)

Platykurtic (Kurtosis < 3)

Sl_No = -1.2 - **Platykurtic**

Ssc_p = -0.60751 - **Platykurtic**

Hsc_p = 0.086901 - **Platykurtic**

Degree_p = -0.09749 - **Platykurtic**

Etest_p = -1.08858 - **Platykurtic**

Mba_p = -0.470723 - **Platykurtic**

Salary = -0.239837 - **Platykurtic**